

STAT

What do you mean by Measures of Central Tendency

→ A measure of central tendency is a typical value around which others figure concentrate.

Types:

1. Arithmetic mean / Average ~~for~~ ^{SR & Exam}
2. Geometric mean.
3. Harmonic mean (SR)
4. Median (SR)
5. Mode (SR)

1] Arithmetic mean: ~~for~~

for ungroup data;

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n}$$
$$= \frac{\sum_{i=1}^n x_i}{n}$$

for group data;

$$\bar{x} = \frac{\sum_{i=1}^n f_i x_i}{\sum_{i=1}^n f_i}$$

Calculation of Median (grouped data).

$$\text{Median} = L + \frac{\frac{n}{2} - F}{f} \times C$$

where,

L = Lower limit of the median class.

n = Total number of observation.

F = Cumulative frequency of the class just preceding

f = frequency.

C = length.

* Median [for Group data] ***

Number of Classes	frequency (f_i)	Cumulative frequency (F_c)
150 - 155	5	5
155 - 160	2	7
160 - 165	6	13
165 - 170	8	21 F_c
<u>170 - 175</u>	<u>9 $\rightarrow f_m$</u>	<u>30</u>
175 - 180	11	41
180 - 185	6	47
185 - 190	3	50
	$\Sigma f_i = 50$	

L
Median
Class

$$n = 50$$

$$\frac{n}{2} = 25$$

C =

Step 1: Median શ્રેણી શોધવા માટે Cumulative frequency
કલ્પના કરવામાં આવે છે.

Step 2: $\frac{n}{2}$ નું મૂલ્ય શોધવામાં આવે છે.

Step 3: $\frac{n}{2}$ નું મૂલ્ય Cumulative frequency સાથે તુલના કરવામાં આવે છે.
જ્યાં સુધી Cumulative frequency $\frac{n}{2}$ થી વધુ થાય છે, તેને Median Class કહેવાય.

Step 4: Median Class નું Lower Limit શ્રેણી L માં મૂકવામાં આવે છે.

Step 5: Median Class નું frequency શ્રેણી f_m માં મૂકવામાં આવે છે.

Step 6: C શ્રેણી નો મૂલ્ય શોધવામાં આવે છે.

Step 7: F_c શ્રેણી Cumulative frequency શ્રેણી નો મૂલ્ય શોધવામાં આવે છે.

$$\therefore \text{Median} = L + \frac{\frac{n}{2} - F_c}{f_m} \times C$$

$$= 170 + \frac{25 - 21}{9} \times 5$$

$$= 172.22$$

Mode: [group data] ~~11~~

$$= L + \frac{\Delta_1}{\Delta_1 + \Delta_2} \times C$$

←

Number of gates	Frequency (f_i)
150 - 155	5
155 - 160	2
160 - 165	7 Δ_1
165 - 170	10 [Maximum frequency]
170 - 175	9 Δ_2
175 - 180	8
180 - 185	6
185 - 190	3

Lower limit or, L

C, length
5

For group data,

$$\begin{aligned} \text{Mode} &= L + \frac{\Delta_2}{\Delta_1 + \Delta_2} \times C \\ &= 165 + \frac{3}{3+1} \times 5 \\ &= 168.75 \text{ mm.} \end{aligned}$$

← Calculate Average Profit;

C	frequency	X_i	$f_i X_i$
0-10	10	5	50
10-20	20	15	300
20-30	35	25	875
30-40	40	35	1400
40-50	25	45	1125
50-60	25	55	1375
60-70	15	65	975 975
$\Sigma f_i = 120$		$\Sigma f_i X_i = 6100$	

$$\begin{aligned}
 \therefore AM = \bar{x} &= \frac{\Sigma f_i X_i}{\Sigma f_i} \\
 &= \frac{6100}{120} \\
 &= 35.88.
 \end{aligned}$$

Other Measures

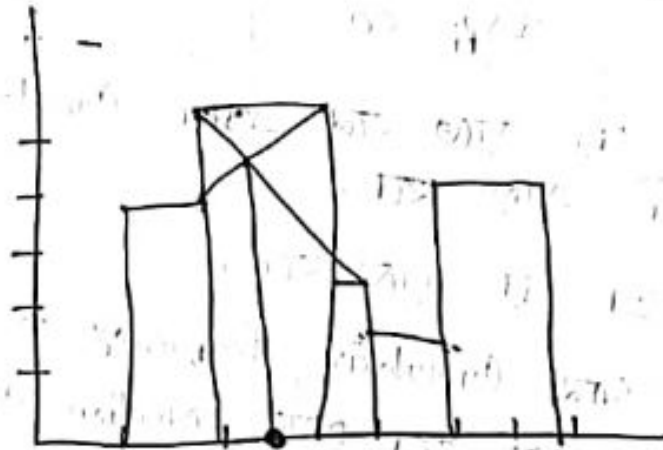
Median এর মূল বস্তু হলো $\frac{n}{2}$ এর সমান হবে,

Quartiles, $Q_i = L + \frac{\frac{ixn}{4} - F_c}{f_a} \times C$; $i = 1, 2, 3$.

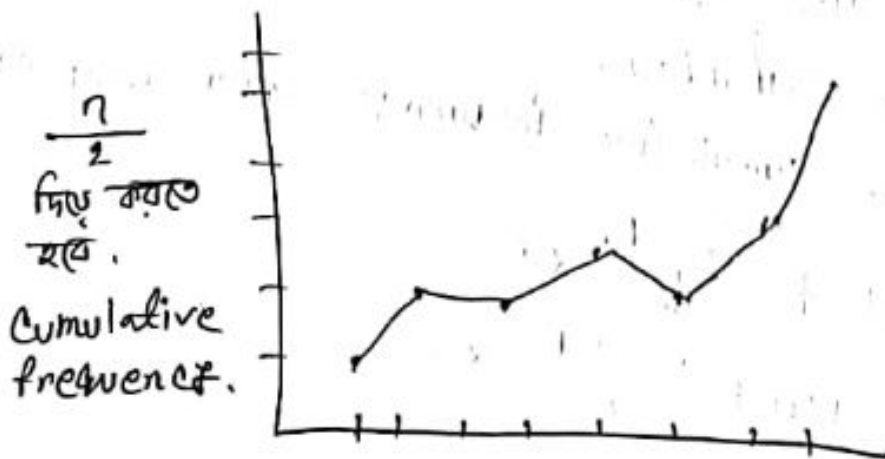
Deciles, $D_i = L + \frac{\frac{ixn}{10} - F_c}{f_a} \times C$; $i = 1, 2, 3, \dots, 9$.

Percentiles, $P_i = L + \frac{\frac{ixn}{100} - F_c}{f_p} \times C$; $i = 1, 2, 3, \dots, 99$.

← Locate mode graphically; 

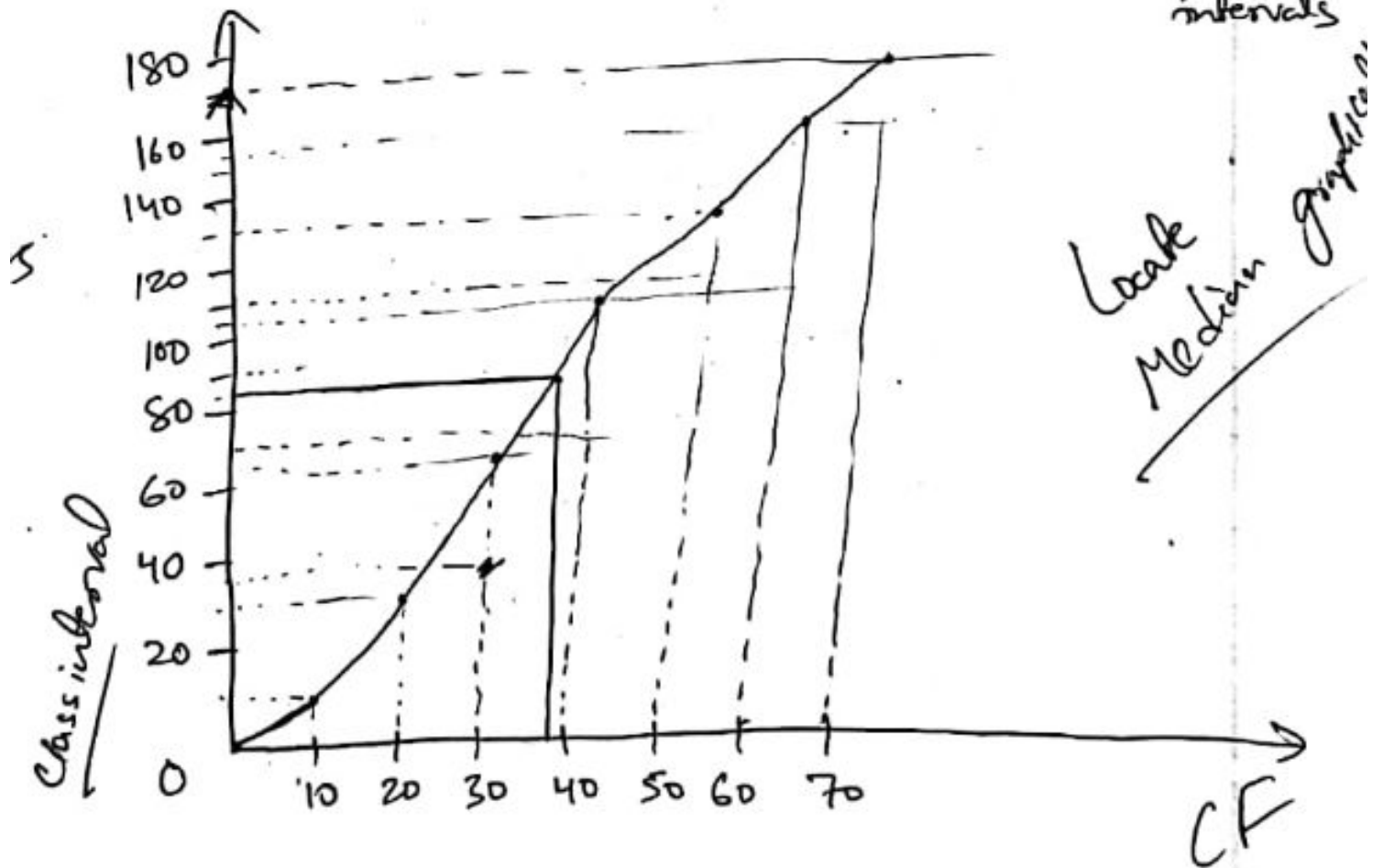
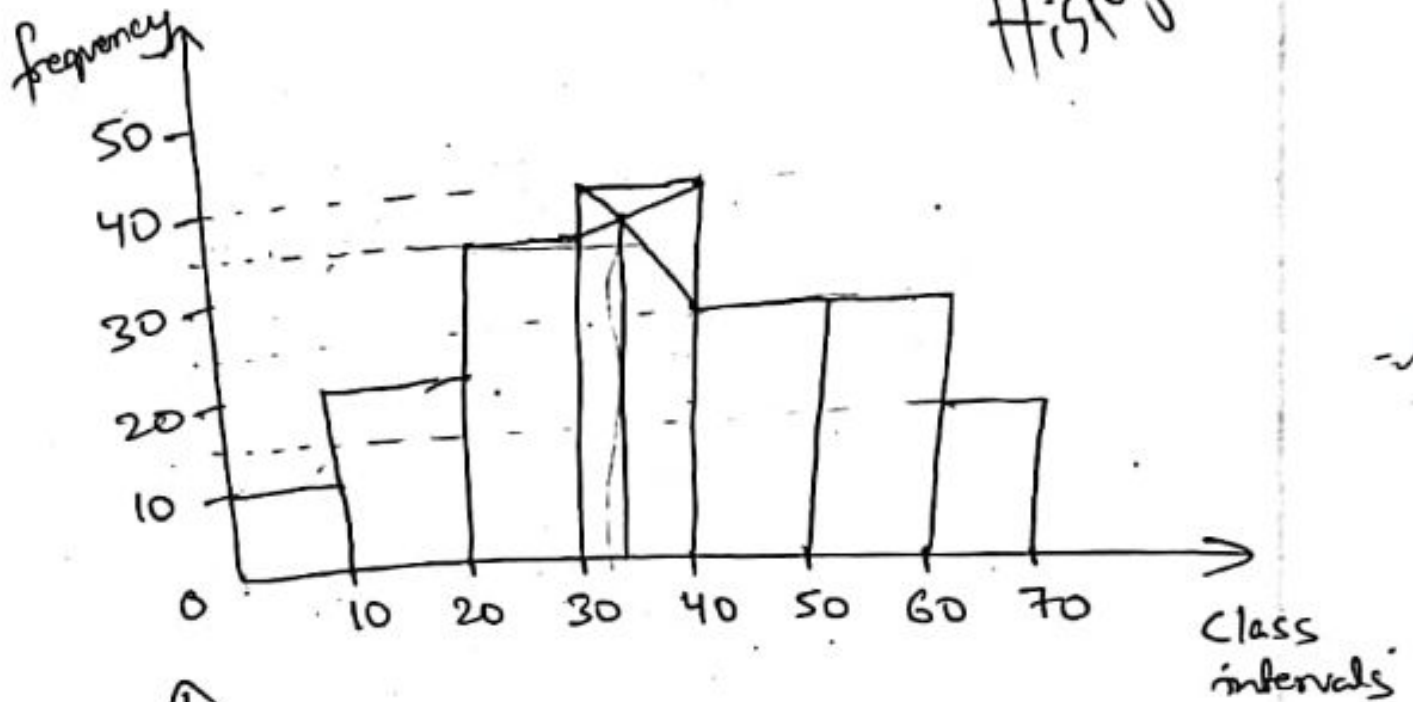


← Locate median graphically; 



iii) Locate Mode Graphically

Histogram



Locate Median Graphically